

ANALYSIS OF SWORDFISH (*XIPHIAS GLADIUS*) CATCH RATES IN THE CENTRAL-EASTERN MEDITERRANEAN

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SUMMARY

Indices of abundance of swordfish (Xiphias gladius) from the Greek longline fisheries operating in the eastern Mediterranean and the Sicilian longline and gillnet fisheries exploiting the Tyrrhenian Sea and the Straits of Sicily are presented for the period 1987-2009. Annual standardized indices were estimated by means of Generalized Linear Modeling techniques and the predictor variables included the Year, Month and Area of fishing. Results did not demonstrate the presence of any particular trend over time, while there are significant CPUE differences among months and areas in all fisheries.

RÉSUMÉ

Les indices d'abondance de l'espadon (Xiphias gladius) provenant des pêcheries palangrières grecques opérant dans l'Est de la Méditerranée et des pêcheries siciliennes opérant à la palangre et au filet maillant dans la mer Tyrrhénienne et le détroit de Sicile sont présentés pour la période comprise entre 1987 et 2009. Les indices annuels standardisés ont été estimés au moyen des techniques de modélisation linéaire généralisée et les variables de prédiction comprenaient l'année, le mois et la zone de pêche. Les résultats n'ont pas apparaître de tendance particulière au cours du temps, bien que des différences significatives de CPUE existent entre les mois et les zones de toutes les pêcheries.

RESUMEN

Se presentan, para el periodo 1987-2009, los índices de abundancia de pez espada (Xiphias gladius) de las pesquerías de palangre griegas que operan en el Mediterráneo oriental y de las pesquerías de red de enmalle y palangre sicilianas que faenan en el mar Tirreno y en el Estrecho de Sicilia. Los índices estandarizados anuales se estimaron por medio de técnicas de modelación lineales generalizadas y las variables independientes incluían Año, Mes y Área de pesca. Los resultados no mostraron la presencia de ninguna tendencia particular en el tiempo, mientras que existen diferencias significativas en la CPUE entre los meses y áreas en todas las pesquerías.

KEYWORD

Swordfish, Mediterranean, Catch/effort

1. Introduction

Swordfish (*Xiphias gladius*) is a commercially important migratory fish heavily fished in the Atlantic and Mediterranean. Greece and Italy are among the most important swordfish producers in the Mediterranean and, in the latest years, account for about 50-60% of the total Mediterranean production. (Anon. 2008)

Greek swordfish fisheries exploit the eastern part of the Mediterranean basin covering a large area, extending from the east Ionian to the Levantine seas. The gear used is drifting surface longlines. Italian fisheries mainly exploit the Adriatic, Ionian and Tyrrhenian seas using surface longlines and gillnets.

The main goal of the present work is to estimate annual standardised abundance indices based on commercial catch per unit effort (CPUE) data series obtained from the most important Greek and Italian fishing fleets. In

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principle, it is attempted to update a previously estimated series (Tserpes et al., 2008) through the use of additional data. Data analysis has been accomplished by means of widely used Generalised Linear Modelling (GLM) techniques.

2. Materials and methods

Data have been collected in the frames of past European and national projects and included spatial and temporal information on catch and effort in as much as possible detail, i.e. on an individual boat trip basis. CPUE observations were expressed in terms of kg/1000 hooks for the longliners and in terms of kg/km for the gillnetters. Sampling, which was based on information from landings on pilot ports, covered the period 1987-2009 and included the main Greek and Italian fleets exploiting different areas of the central and eastern Mediterranean (**Figure 1**).

In the case of Greece, a total of 4213 observations were analyzed. These covered the activities of the two main swordfish fleets operating in the country, the fleets of Kalymnos and Hania. Generally, catches of both fleets represent 50-70% of the total Greek production (Tserpes *et al.*, 2002). These fleets mainly exploit the central, south-eastern Aegean Sea but occasionally extend their activities to the northern Aegean and Levantine basin. Fishing is carried out using drifting surface longlines through February to September while is prohibited by law from October to January. In the last eight years, the traditional long lines have been modified, resembling the ones used for the tuna fishery in the Atlantic. The modified gear, which is known as American-type longline, is set deeper than the traditional one and uses fluorescent material to attract the fish. Past studies have demonstrated catchability differences among the two gear types (Tserpes and Peristeraki, 2004); thus the “gear” type has been taken into account in the analysis.

In the case of Italy, a total of 5196 observations were analyzed from the Sicilian longline and gillnet fleets. The Sicilian fleet, which is among the larger swordfish fleets in the Mediterranean, mainly exploits the Tyrrhenian Sea but occasionally expands its activities into a much wider area. The longline fishery mainly operates from August to December while the driftnet fishing season usually lasts from April to August.

CPUE data were analysed, separately by country and fleet, by means of Generalised linear modelling (GLM) techniques (McCullagh and Nelder 1983). Based on the deviance residuals plot, models assuming a Gamma error structure with a log link function were found to be the most appropriate for all examined data sets. The models included year, month, area and gear type (only for the Greek longliners) as main effects, as well as all possible second order interactions between all factors, except “year”. Interaction terms including the “year” effect were not examined as they could bias the year effect standardised estimates.

Thus, the general form of the GLM used was:

$$\text{CPUE} \sim \text{Year} + \text{Month} + \text{Area} + \text{Gear} + \text{Month:Area} + \text{Month:Gear} + \text{Area:Gear}$$

Model fitting was accomplished under the R language environment (R Development Core Team, 2008) and statistical inference was based on the 95% confidence level.

3. Results and discussion

3.1 Eastern Mediterranean (Greek longliners)

A total of 4213 data records were analysed that were collected in the period 1987-2009 with the exception of 1989, 1996 and 1997. Five fishing areas were considered: A = Cretan sea, B = Central Aegean, C = South-eastern Aegean, D = Levantine and E = North Aegean (see also **Figure 1**). There is no any outstanding feature in the deviance residual plot suggesting that the model is inappropriate for the observations (**Figure 2**). The analysis of deviance table indicated that the model explained about 25% of the total variation. All effects were significant on the 95% statistical level (**Table 1**).

The effect of the significant predictors on CPUE is shown on the y-axis for different values of the predictor (x-axis) (**Figure 3**). Although CPUE levels do not show any particular trend over years, it seems that rates in the last decade are generally lower than those estimated for the previous years. Estimated indices are given in **Table 3**. The american-type longline has higher catch rates that the conventional one in all areas and months, while the monthly pattern of catch rates differ among areas and gears (**Figure 3**).

3.2 Central Mediterranean (Sicilian longliners)

A total of 3321 data records were analysed that were collected throughout the year from 1991 to 2009, with the exception of 1993 and 1996. Three fishing areas were considered: I = Straits of Sicily, J = South Tyrrhenian and K = Central Tyrrhenian seas (see also **Figure 1**). The model provided a good fit to the data as it was demonstrated by the deviance residual plot (**Figure 4**). The analysis of deviance table indicated that it explained about 16% of the total variation. All factors were significant on the 95% statistical level (**Table 2**).

The effect of the significant predictors on CPUE is shown on the y-axis for different values of the predictor (x-axis) (**Figure 5**). Similarly to the Greek longliners, the CPUE levels do not show any particular trend among years. Standardised annual indices are given in **Table 3**.

3.3 Central Mediterranean (Sicilian gillnetters)

A total of 1875 data records were analysed that were collected throughout the year from 1991 to 2009, with the exception of 1993 and 1996. Three fishing areas were considered: I = Straits of Sicily, J = South Tyrrhenian and K = Central Tyrrhenian seas (see also **Figure 1**). The model provided a good fit to the data as it was demonstrated by the deviance residual plot (**Figure 4**). The analysis of deviance table indicated that it explained about 65% of the total variation. All factors were significant on the 95% statistical level (**Table 2**).

The effect of the significant predictors on CPUE is shown on the y-axis for different values of the predictor (x-axis) (**Figure 5**). Similarly to the rates estimated for the previous fisheries CPUE does not any particular overall trend. Standardised annual indices are given in **Table 3**.

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Table 1. Analysis of deviance table for the Gamma-based GLM model fitted to longline CPUE data from the Greek fleets.

Source of variation	Df	Deviance	Resid. Df	Resid. Dev	F	Pr(>F)
NULL			4212.00	1936.13		
year	19	339.65	4193.00	1596.48	42.61	<0.001
Gear	1	22.71	4192.00	1573.77	54.14	<0.001
month	7	20.75	4185.00	1553.02	7.07	<0.001
Area	4	35.62	4181.00	1517.40	21.23	<0.001
month:Area	28	33.00	4153.00	1484.40	2.81	<0.001
Gear:month	7	27.28	4146.00	1457.12	9.29	<0.001
Gear:Area	4	13.66	4142.00	1443.46	8.14	<0.001

Table 2. Analysis of deviance table for the Gamma-based GLM model fitted to longline CPUE data from the Sicilian fleet.

Source of variation	Df	Deviance	Resid. Df	Resid. Dev	F	Pr(>F)
NULL			3320.00	2239.57		
year	16	148.85	3304.00	2090.72	16.59	<0.001
month	11	61.37	3293.00	2029.35	9.95	<0.001

Table 3. Analysis of deviance table for the Gamma-based GLM model fitted to gillnet CPUE data from the Sicilian fleet.

Source of variation	Df	Deviance	Resid. Df	Resid. Dev	F	Pr(>F)
NULL			1874.00	2795.14		
year	16	1462.23	1858.00	1332.91	164.16	<0.001
month	9	182.18	1849.00	1150.72	36.36	<0.001

Table 4. Standardized abundance indices by year and fleet. Indices are expressed in terms of kg/1000hooks for the longliners (LL) and kg/km for the gillnetters (GN).

Year	GR LL	IT LL	IT GN
-1987	142.37		
1988	182.89		
1989			
1990	152.99		8.31
1991	205.78	94.81	9.80
1992	77.39	92.80	16.87
1993	141.47		13.04
1994	200.65	100.23	9.49
1995	117.24	127.30	14.65
1996			9.33
1997		97.37	14.04
1998	245.18	127.95	10.12
1999	193.20	161.64	12.71
2000	154.22	90.04	14.92
2001	161.25	141.65	13.06
2002	127.74	203.21	
2003	164.71	155.55	
2004	155.66	103.38	15.15
2005	157.53	114.58	12.07
2006	164.23	138.63	30.74
2007	171.59	118.03	0.00
2008	160.35	123.49	3.29

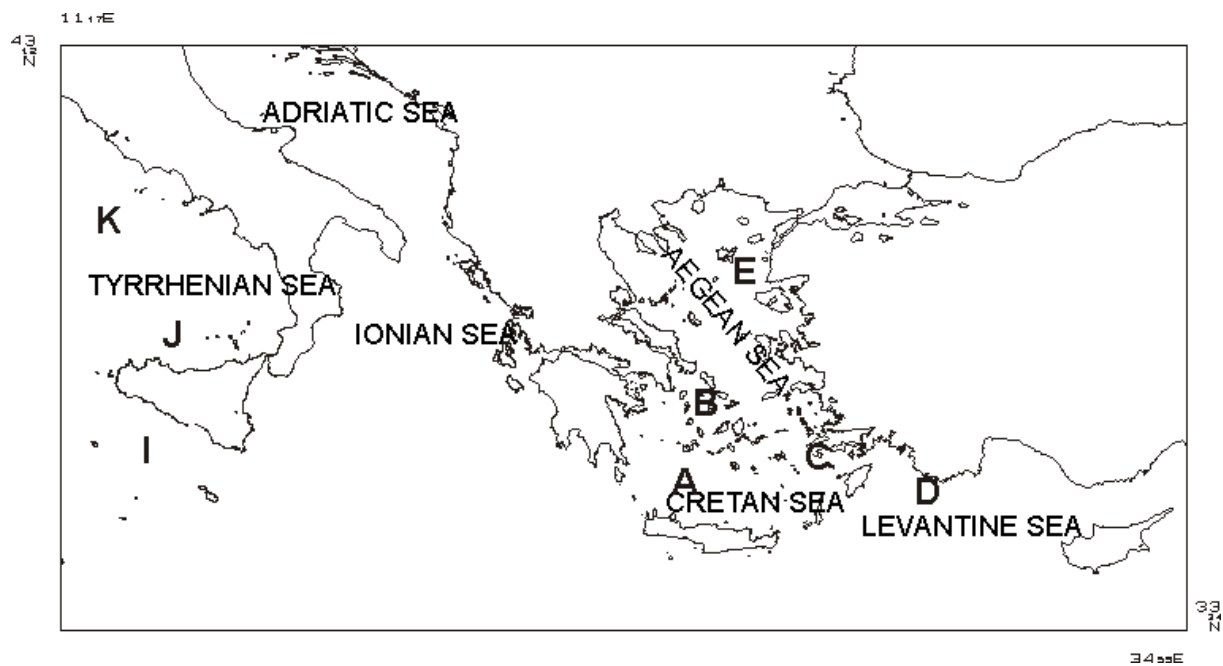


Figure 1. Map of the central-eastern Mediterranean indicating the main areas exploited by the studied fleets. A = Cretan Sea, B = Central Aegean, C = Southeastern Aegean, D = Levantine, E = North Aegean, I = Straits of Sicily, J = Southern Tyrrhenian and K = Central Tyrrhenian.

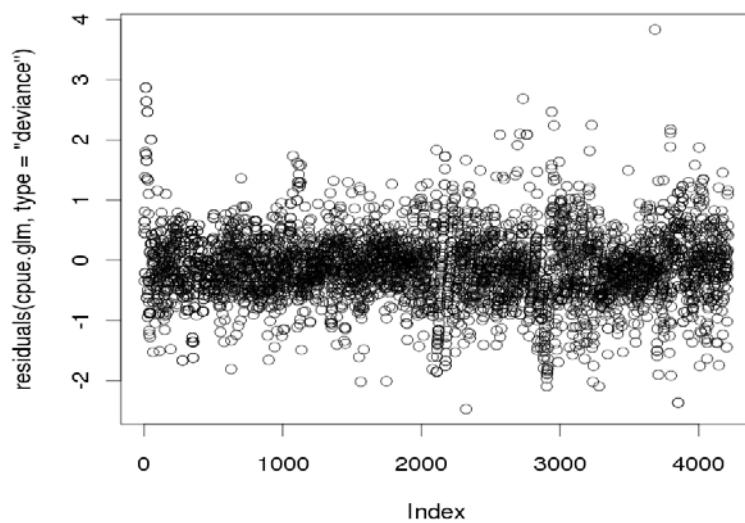


Figure 2. Residual deviance of the generalized linear model fitted to Greek longline data.

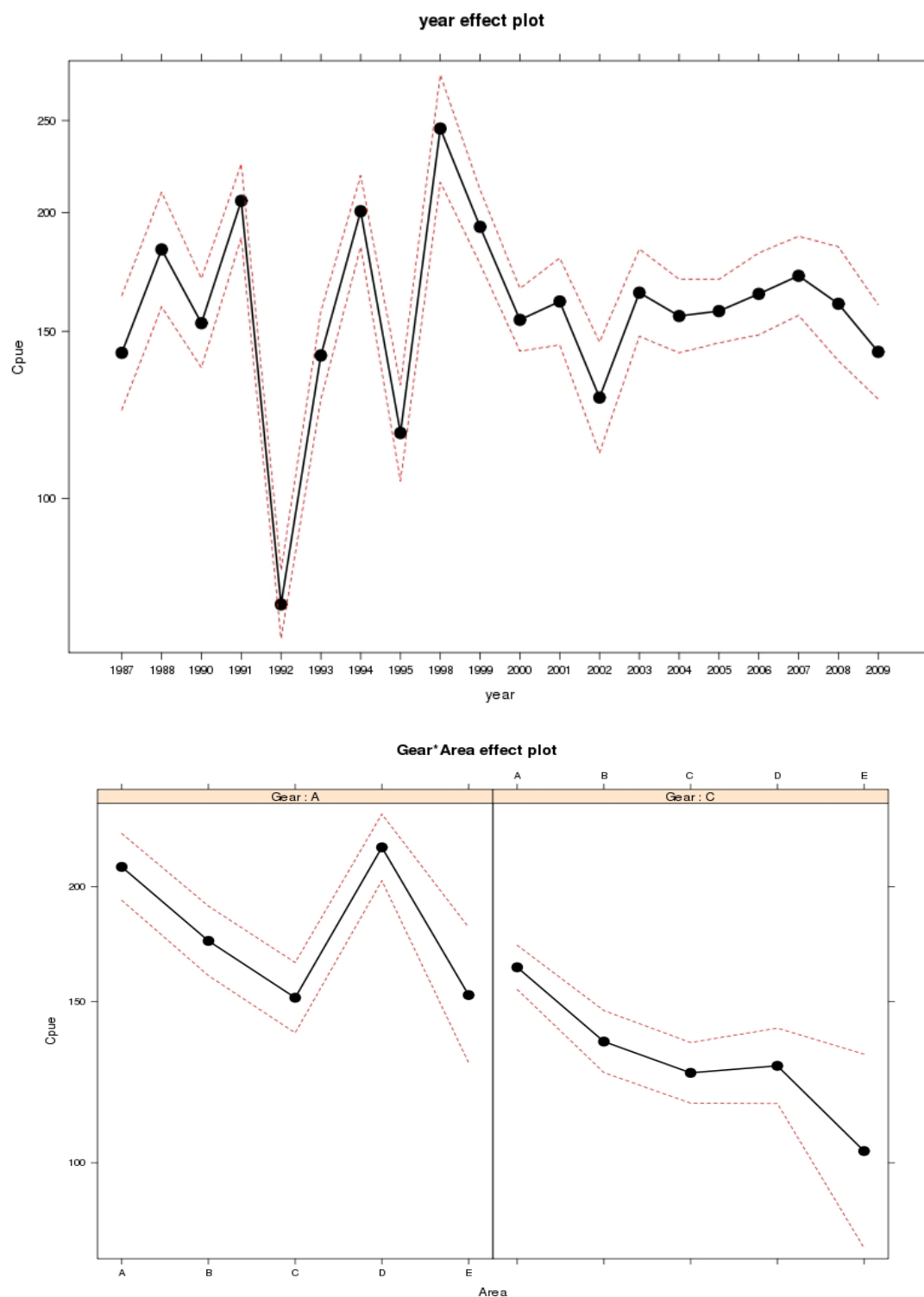


Figure 3. Generalized linear model derived significant effects on CPUE index for the Greek longline data. Each plot represents the contribution of the corresponding variable to the fitted predictor. Broken lines indicate two standard errors.

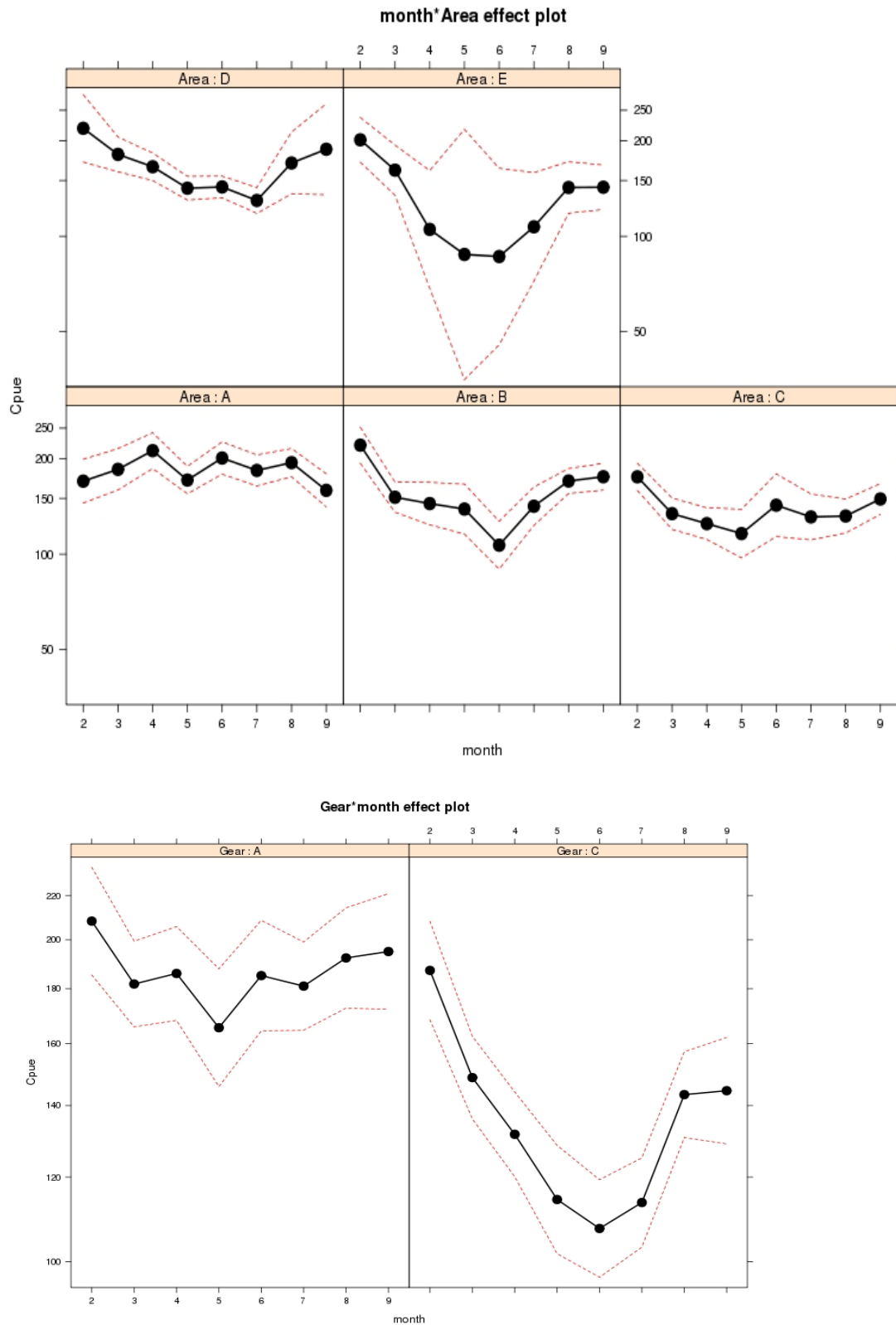


Figure 3. (continued)

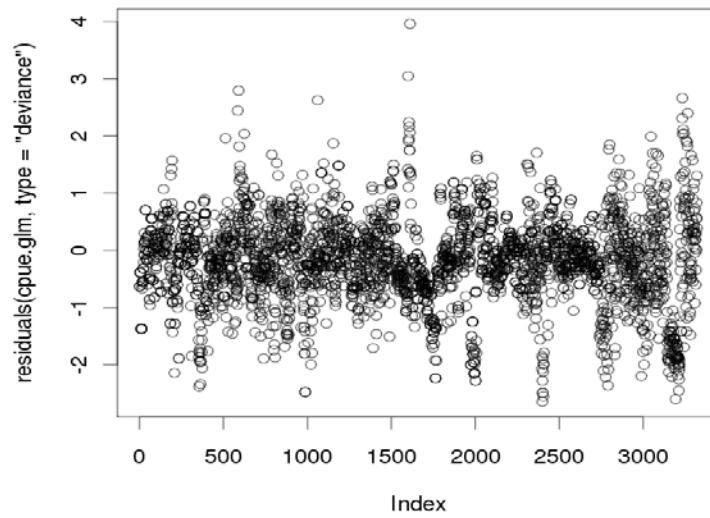


Figure 4. Residual deviance of the generalized linear model fitted to Sicilian longline data.

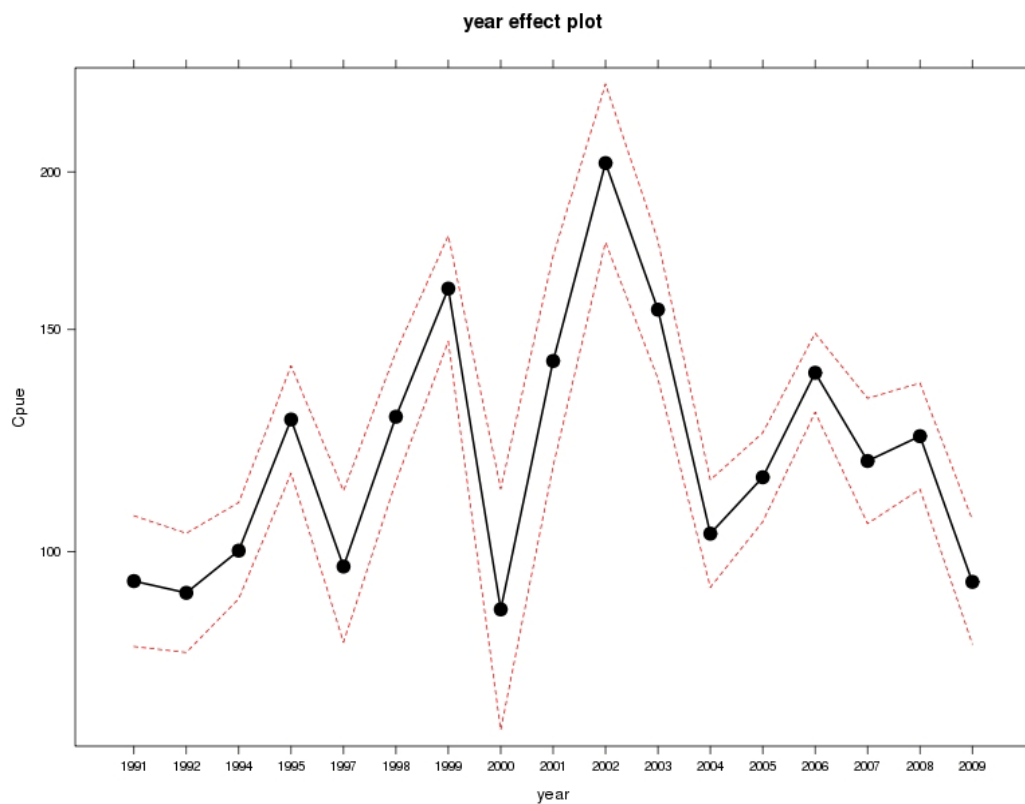


Figure 5. Generalized linear model derived significant effects on CPUE index for the Sicilian longline data. Each plot represents the contribution of the corresponding variable to the fitted predictor. Broken lines indicate two standard errors.

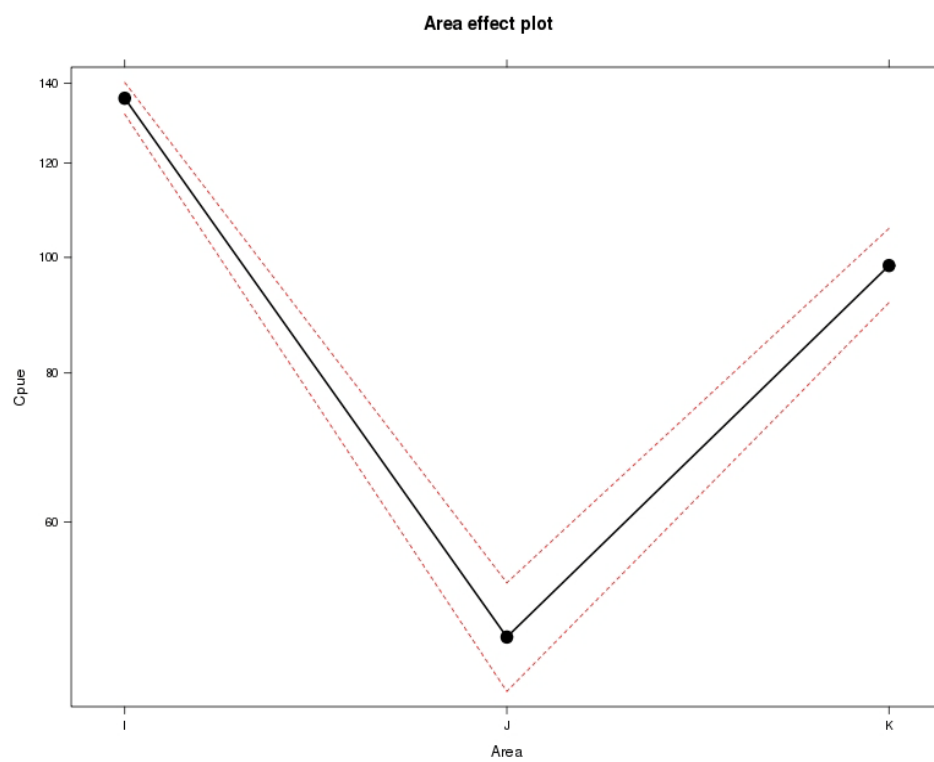
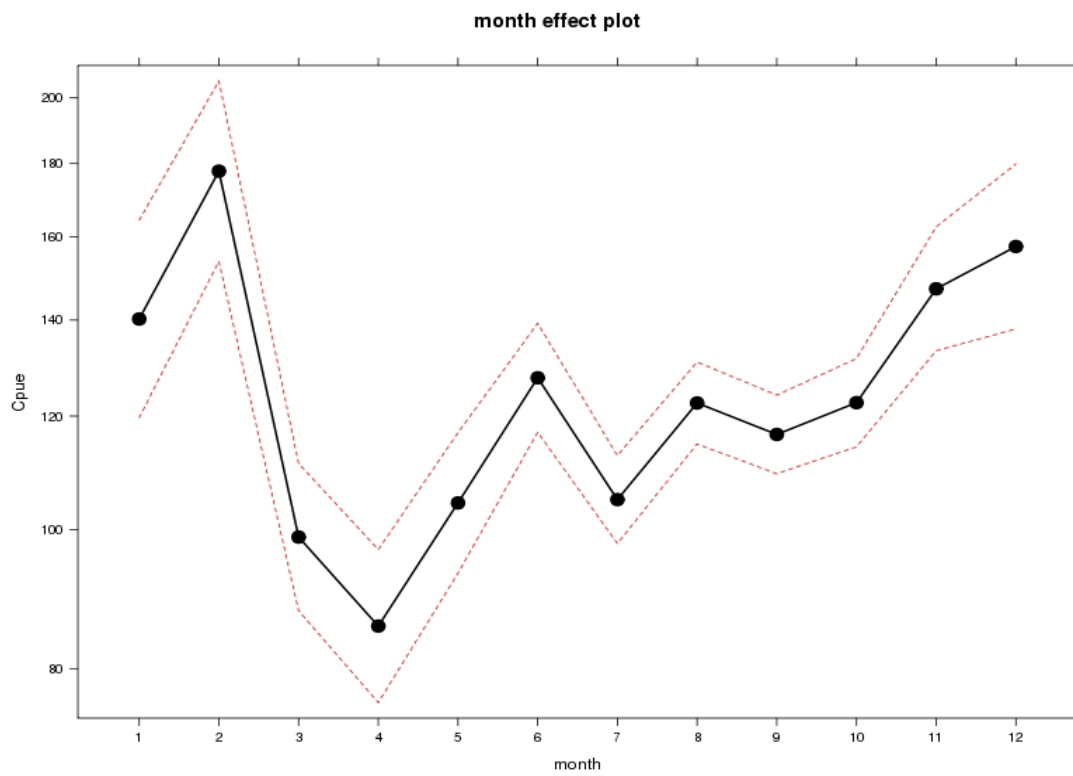


Figure 5. (continued)

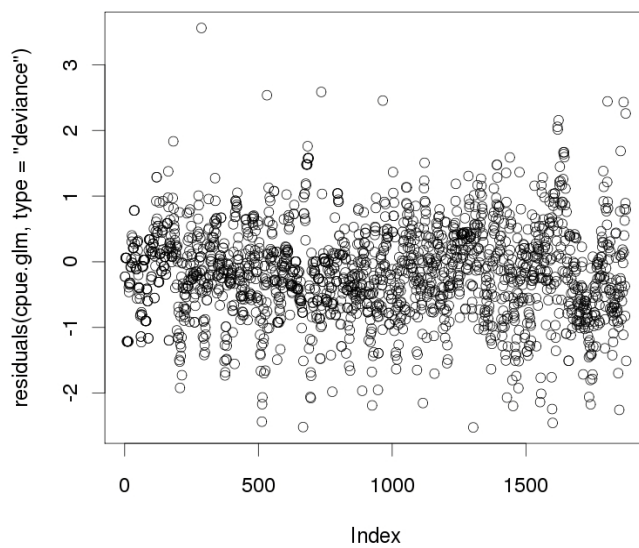


Figure 6. Residual deviance of the generalized linear model fitted to Sicilian gillnet data.

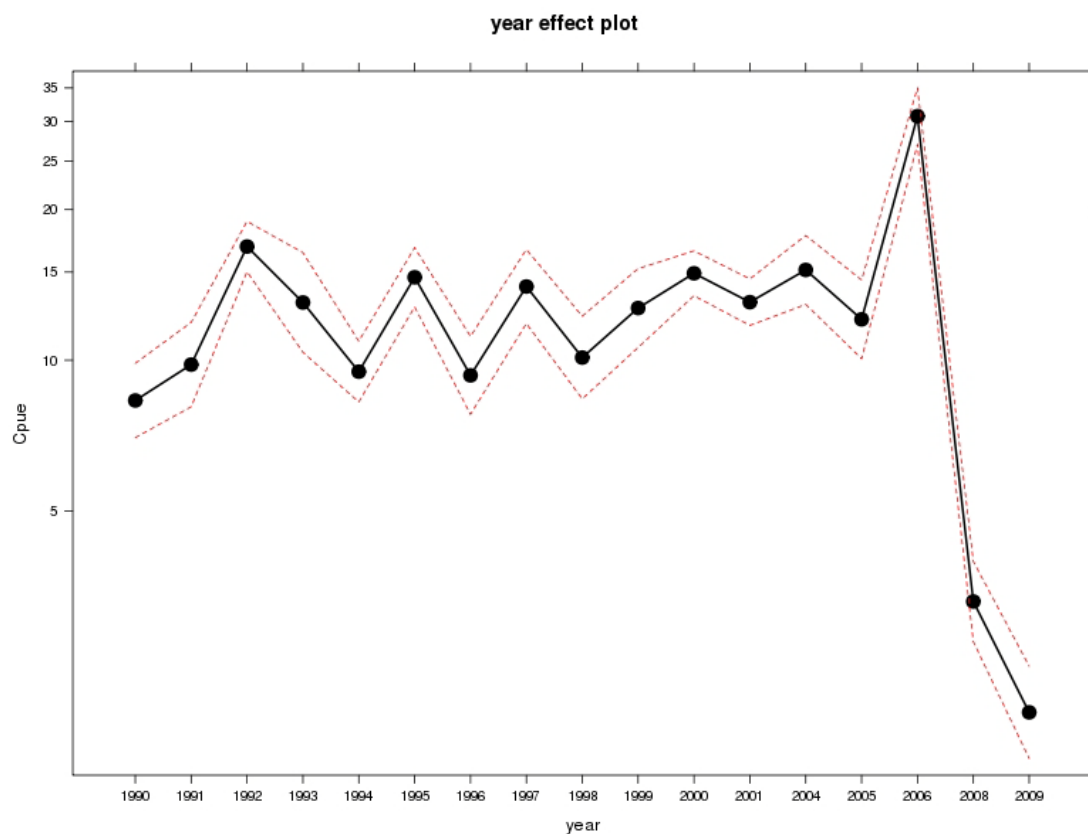


Figure 7. Generalized linear model derived significant effects on CPUE index for the Sicilian gillnet data. Each plot represents the contribution of the corresponding variable to the fitted predictor. Broken lines indicate two standard errors.

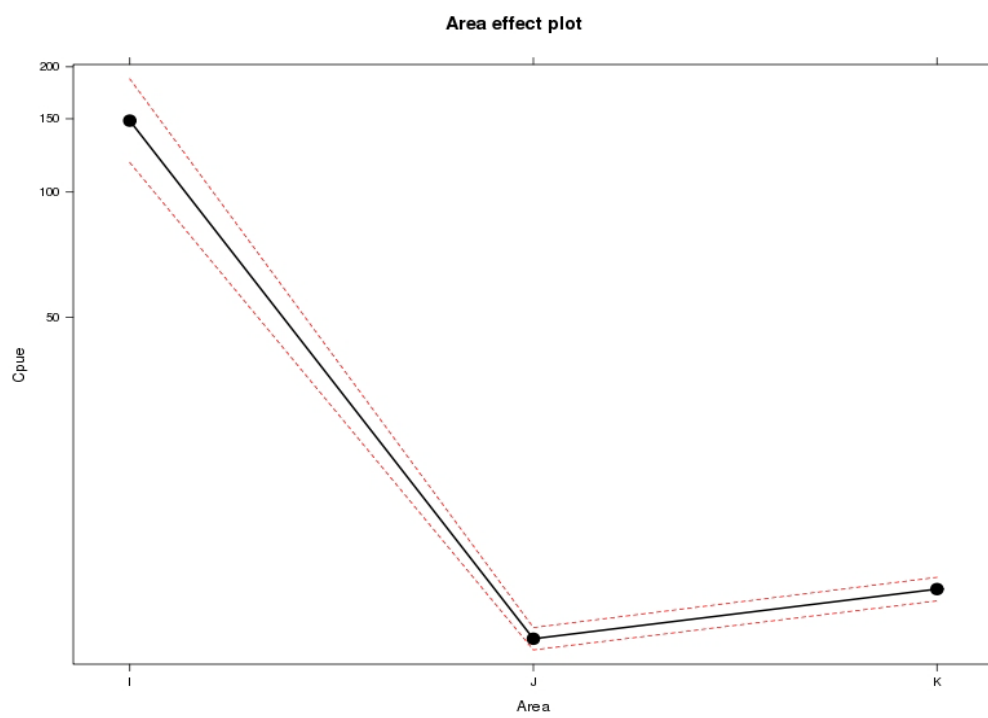
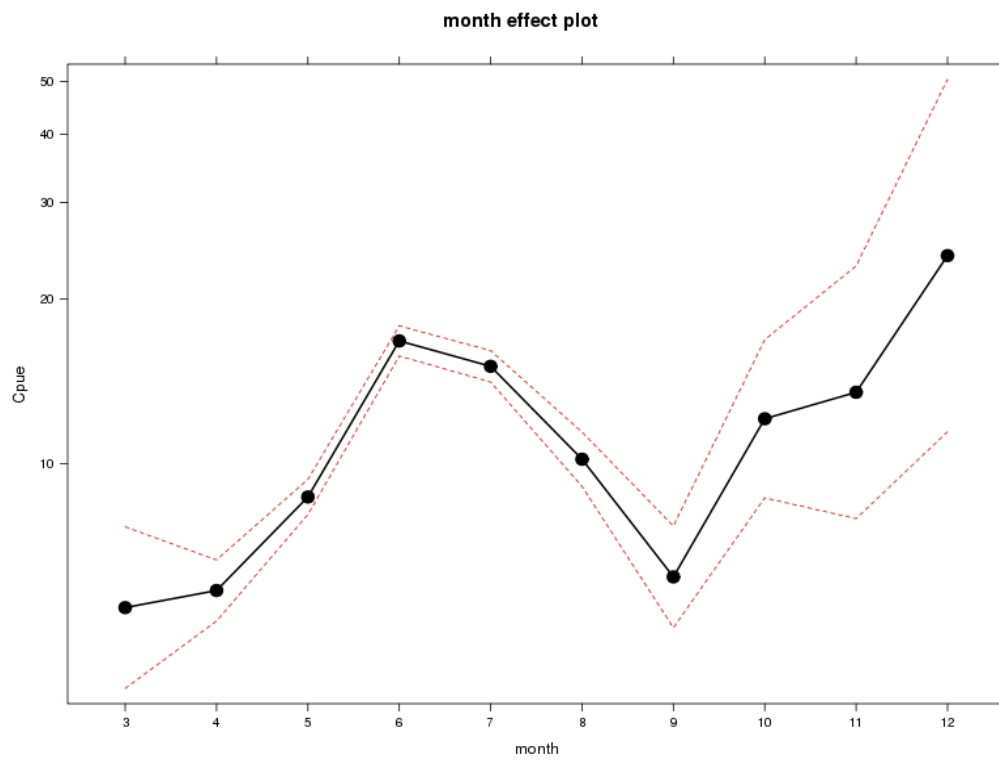


Figure 7. (continued)